

# Deep Reinforcement Learning

## *Advanced Algorithms*

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# Content

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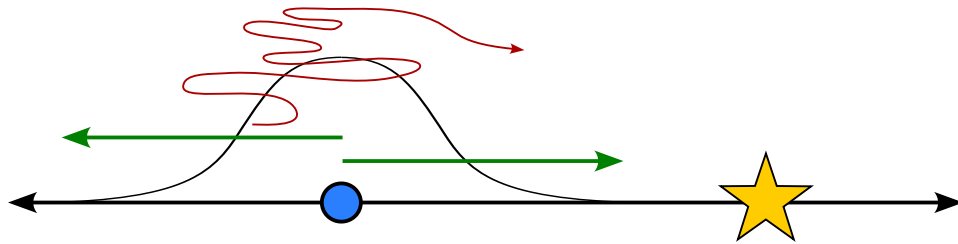
- TRPO / PPO
- DDPG / TD3 / SAC
- Dueling DQN?

# Where are we?

| Chapter | Topic                                                        | Content                                       |
|---------|--------------------------------------------------------------|-----------------------------------------------|
|         | <b>Basics &amp; tabular methods</b>                          |                                               |
| 1-5     | Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning | RL basics in finite dimensions                |
|         | <b>Deep-learning-based methods</b>                           |                                               |
| 6       | Brief introduction to deep learning                          | The basics for what comes next                |
| 7       | Value function approximation                                 | Value estimation with function approximation  |
| 8       | Deep Q-learning                                              | Q-learning with neural networks               |
| 9       | Policy gradients                                             | Direct optimization of the policy             |
| 10      | Actor-critic algorithms                                      | Improved policy gradients via value functions |
| 11      | <b>Advanced algorithms</b>                                   |                                               |
|         | <b>Model-Based Control</b>                                   |                                               |
|         | <b>Advanced Topics</b>                                       |                                               |

# Natural / covariant policy gradients (TRPO)

# Ill-conditioned gradients



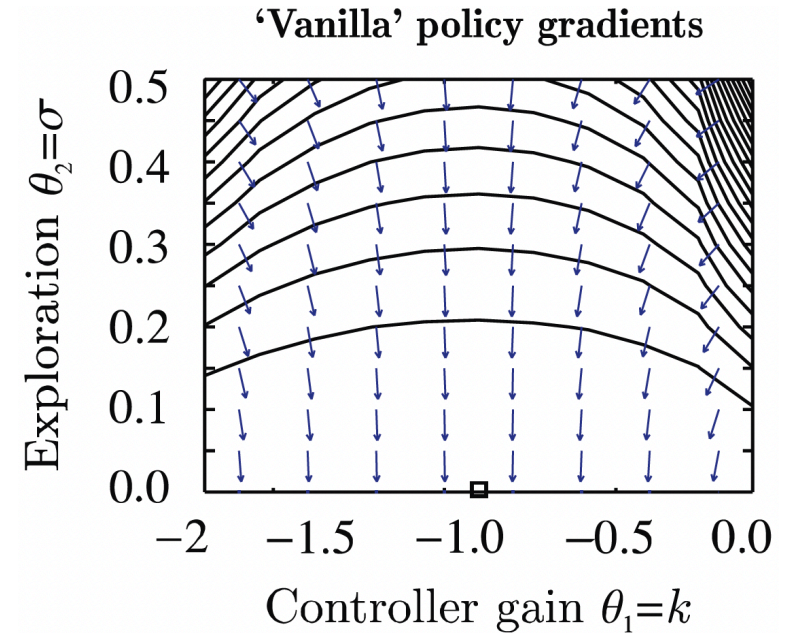
Inspired by Sergey Levine's [CS285 lecture](#).

$$r_t = -s_t^2 - a_t^2$$

$$\log \pi_{\phi}(a_t|s_t) = -\frac{1}{2\sigma^2}(ks_t - a_t)^2 + C, \quad \phi = (k, \sigma).$$

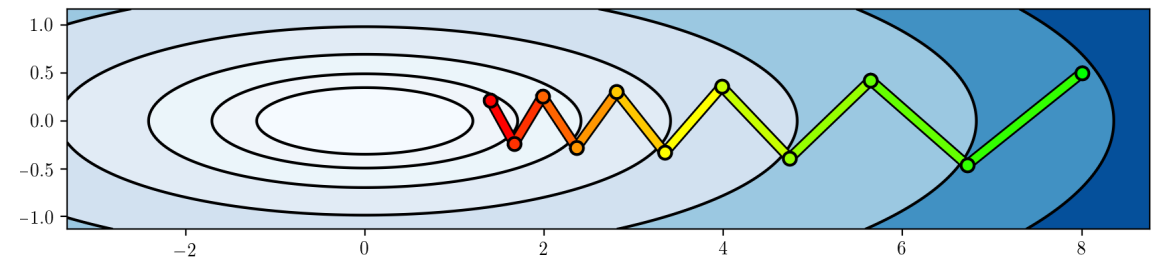
$$\nabla_{\phi} \log \pi_{\phi}(a_t|s_t) = \begin{pmatrix} -\frac{(ks_t - a_t)s_t}{\sigma^2} \\ \frac{(ks_t - a_t)^2}{\sigma^3} \end{pmatrix}$$

- **Optimum:**  $\phi^* = (-1, 0)$ .
- **Issue:** The gradients do not point towards the optimum for smaller  $\sigma$ !
  - The  $\sigma^{-3}$  term blows up.



Source: [\(Peters and Schaal 2008\)](#)

Closely related to ill-conditioned optimization problems:





# References

Peters, Jan, and Stefan Schaal. 2008. "Natural Actor-Critic." *Neurocomputing* 71 (7): 1180–90.